IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

SPEECH PROCESSING

Monday 22 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Speech Processing

1. Consider a discrete-time speech signal $s(n)$ corrupted by additive noise $\nu(n)$. The noisy speech signal can be written in the time domain as $y(n) = s(n) + \nu(n)$.

   a) Derive in full detail the Weiner filter that could be used to reduce the noise and state under what conditions the Weiner filter is optimal. [4]

   b) In a wideband telecommunication system, a noisy speech signal $y(n)$ is obtained from a single microphone and in the frequency domain can be written as

   $$Y(l,k) = S(l,k) + V(l,k)$$

   for time-frame index l and frequency index k.

      i) Explain what is meant by spectral variance, $\phi_y(l,k)$, of the noisy speech signal. [2]

      ii) Write down the mathematical definition for $\phi_y(l,k)$. [4]

      iii) Describe in detail a method based on Voice Activity Detection, and a method based on Minimum Statistics, to compute an estimate of the spectral variance of the noise, $\hat{\phi}_v(l,k)$. Include mathematical expressions for the estimators in your description where relevant. [4]

      iv) Compare and contrast the advantages and disadvantages of these alternative methods. [2]

   c) Consider a noise reduction algorithm applied to the noisy speech signal. The device uses the method of amplitude spectral subtraction.

   Show that the output signal of the noise reduction algorithm at time-frame l and frequency index k can be written

   $$Z(l,k) = H(l,k)Y(l,k)$$

   and write down an expression for $H(l,k)$ in terms of $\hat{\phi}_v(l,k)$. [4]

2. a) In evaluating a speech recognition system, state the 3 types of word-errors that are considered when computing the word error rate. Explain each type of error, and give the formula for the word error rate. [4]

b) Consider a speech recognition system based on Deep Neural Networks (DNN). Data augmentation is a common strategy to increase the quantity of training data, and improve the robustness of the trained DNN models.

i) State 3 ways in which speech signals can be modified to achieve augmentation of the training data. [3]

ii) Explain why such augmentation can improve the performance of a speech recognition system. [2]

c) Consider a speech recognition system based on Hidden Markov models (HMM).

i) For an arbitrary state, s, in the HMM, let the probability of a transition from state s to the next state be denoted p. Taking into account only this transition probability p, derive an expression in terms of p for the number of frames aligned to state s. Show all the details of your derivation. [4]

ii) A short speech utterance is represented by feature vectors $\mathbf{x}_1$,$\mathbf{x}_2$,$\mathbf{x}_3$,$\mathbf{x}_4$,$\mathbf{x}_5$. These feature vectors are to be compared with a 3-state HMM for which the output probability densities of the feature vectors for each state are shown in Table 1. The state diagram of the HMM is shown in Figure 2.1, in which the labels on the arrows indicate the state transition probabilities.

Construct the lattice diagram of the model and draw on the lattice all the feasible paths through the lattice. [2]

Determine the maximum probability alignment of the frames to the states of the HMM and the value of B(3,5) corresponding to this alignment. [5]

	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_4$	$\mathbf{x}_5$
State 1	0.5	0.2	0.6	0.4	0.5
State 2	0.5	0.7	0.3	0.1	0.5
State 3	0.5	0.5	0.4	0.6	0.6

Table 1

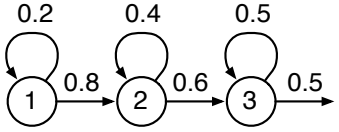


Figure 2.1

3. a) A linear predictor for the speech signal $s(n)$ can be written

$$s(n) = Gu'(n) + \sum_{j=1}^{p} a_j s(n-j).$$

i) Describe how Linear Predictive Coding (LPC) is used in speech signal processing applications. Include any relevant diagrams and mathematical formulae. [5]

ii) Explain what the terms

$$u'(n),\ G,\ \text{and}\ \ a_j$$

represent in this expression. [3]

b) For a linear predictor, and stating any assumptions used, formulate an expression for the prediction error and hence show that the prediction error is minimum when the following expression is satisfied: [3]

$$\sum_{j-1}^{p} a_j \sum_{n\in\{F\}} s(n-j)s(n-i) = \sum_{n\in\{F\}} s(n)s(n-i) \quad \text{for} \quad i = 1,\ \ldots,\ p.$$

Explain the meaning of $\{F\}$. [1]

c) i) Define the complex cepstral coefficients of a speech signal and briefly state their important properties for speech processing. [2]

ii) Derive a recursion to obtain the complex cepstral coefficients from the terms a_j as defined by part a) of this question. [4]

iii) Find the complex cepstral coefficients for the case when

$$[a_0,\ a_1,\ a_2,\ a_3,\ a_4] = [1,\ -0.35,\ -0.25,\ 0.18,\ 0.4].$$

[2]

4. a) Consider a Differential Pulse Code Modulation (DPCM) speech coder.

i) Draw a fully labelled block diagram of a DPCM speech coder. [3]

ii) Describe how each of the blocks in the diagram operates to achieve a reduction in bit rate while maintaining signal quality. [3]

b) Consider a uniform quantizer having quantization intervals of width w. Show that the mean square quantization error is given by

$$\frac{w^2}{12}.$$

State any assumptions used. [4]

c) Consider a uniform quantizer with output levels at values

$$\pm 64(2k-1) \text{ for } k = 1, \ldots, 128.$$

i) Using the formula from above (question part b) calculate the mean square quantization error for this quantizer. [2]

ii) Find the mean square value of a uniformly distributed signal occupying the range of amplitude ± 11200. [2]

iii) Hence, calculate the signal-to-quantization-noise ratio in dB. [2]

d) i) Non-uniform quantization schemes may be advantageous for some signals. Briefly explain why and state the relevant signal characteristics. [2]

ii) Now consider a particular non-uniform quantizer with output levels at values given by

$$\pm\left(2^e(2m+33)-33\right), \quad m = 0, 1, \ldots, 15 \text{ and } e = 0, 1, \ldots 7.$$

Find the smallest and the largest quantization step. [2]